OPEN ENDED LAB (OEL) REPORT: CLO1 & CLO2

ECG Biomedical Signal Processing & Heart Rate Variability (HRV) Analysis System

Course Title: Biomedical Signal Processing	**Student Name:** Biomedical Student
Lab Task: Open-Ended Lab Evaluation	**Roll/ID Number:** BME-2026-09
Department: Biomedical Engineering	**Supervisor:** Dr. Eleanor Vance
Date of Experiment: May 31, 2026	**Evaluation Score:** ______ / ______

1. Laboratory Objectives

1. Implement and validate a digital biomedical signal processing pipeline for high-frequency noise and baseline wander removal from raw electrocardiogram (ECG) data.
2. Establish R-peak detection accuracy using the adaptive Pan-Tompkins derivative-threshold algorithm.
3. Develop an outlier-rejection engine using localized median thresholding to correct ectopic cardiac beats.
4. Compute and interpret linear (time-domain, frequency-domain) and non-linear (Poincaré plot, Sample Entropy) HRV metrics for physiological homeostasis assessment.

2. Physiological Background & Introduction

The autonomic nervous system (ANS) controls the sinus node pacemaker through antagonistic sympathetic (stimulatory) and parasympathetic/vagal (inhibitory) branches. Heart Rate Variability (HRV) is the fluctuation in time intervals between consecutive heartbeats (R-R intervals). Depressed HRV markers serve as diagnostic indicators for autonomic neuropathy, diabetic dysautonomia, stress-induced cardiovascular remodeling, and ischemic risks. Accurate analysis requires high-fidelity digital filtering and R-peak extraction, as small noise remnants or ectopic ventricular contractions (PVCs) drastically distort HRV statistics.

3. Biomedical Filter Design & DSP Methodology

3.1 Baseline Wander Removal (Dual Median Filter)

Baseline drift (0.1–0.5 Hz) caused by respiration is non-linearly tracked using a dual median filter. First, a 200 ms window (removing high-amplitude QRS complexes and T-waves) passes over the ECG. Next, a 600 ms window filters the output to eliminate P-waves. This isolates the baseline wander, which is subtracted from the raw signal. Unlike linear IIR filters, median filtering introduces zero phase delay and leaves QRS wave amplitudes unmodified.

3.2 P-QRS-T Noise Suppression (Butterworth Bandpass)

To capture ECG waves, a 3rd-order Butterworth bandpass filter is designed with a passband of 0.5 to 45.0 Hz. This band suppresses low-frequency breathing drifts and high-frequency 50/60 Hz powerline interference. It is applied using double-pass filtering (scipy.signal.filtfilt), enforcing zero phase shift to prevent shifting QRS peaks in time.

3.3 Moving Enveloping (Pan-Tompkins Algorithm)

The Pan-Tompkins algorithm isolates QRS complexes using a derivative stage, a squaring operation, and a moving window integrator (150ms window size). This integration tracks the energy envelope of the QRS complex. R-peaks are identified using adaptive dual thresholds: the signal threshold adapts to standard beat heights while the noise threshold adjusts to muscular baseline tremors. Refractory constraints (200ms lock-out) prevent T-wave double-triggering, and a back-search threshold scans back if beats are skipped.

4. Signal Processing Pipeline Results & Figures

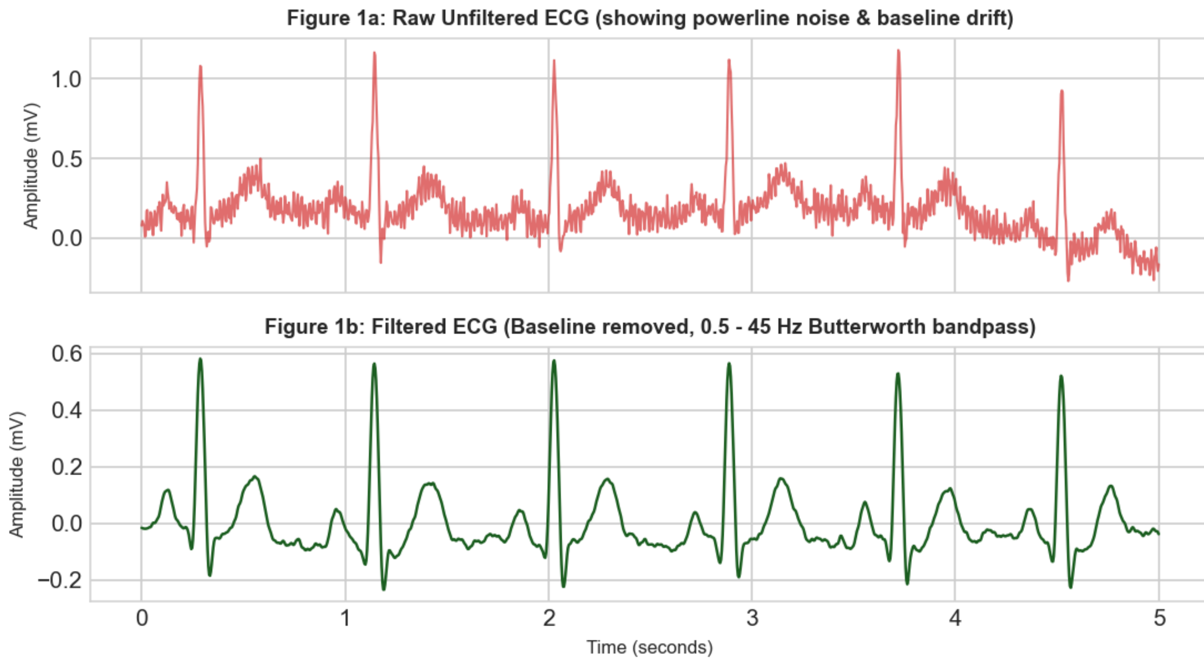

Figure 1: Comparison between raw, noisy ECG and the filtered, baseline-corrected signal.

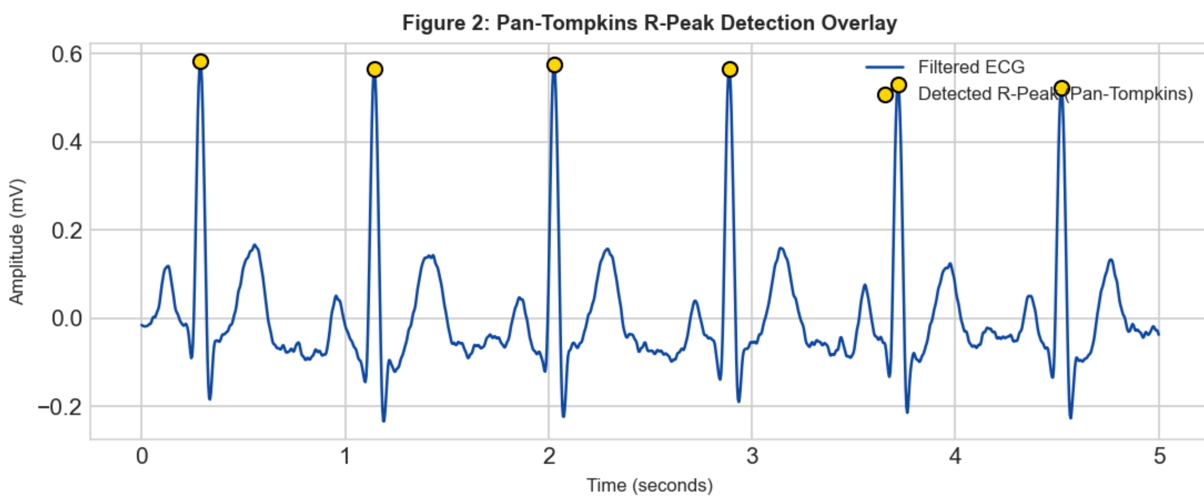

Figure 2: Pan-Tompkins QRS output. Overlay dots represent aligned R-peak locations.

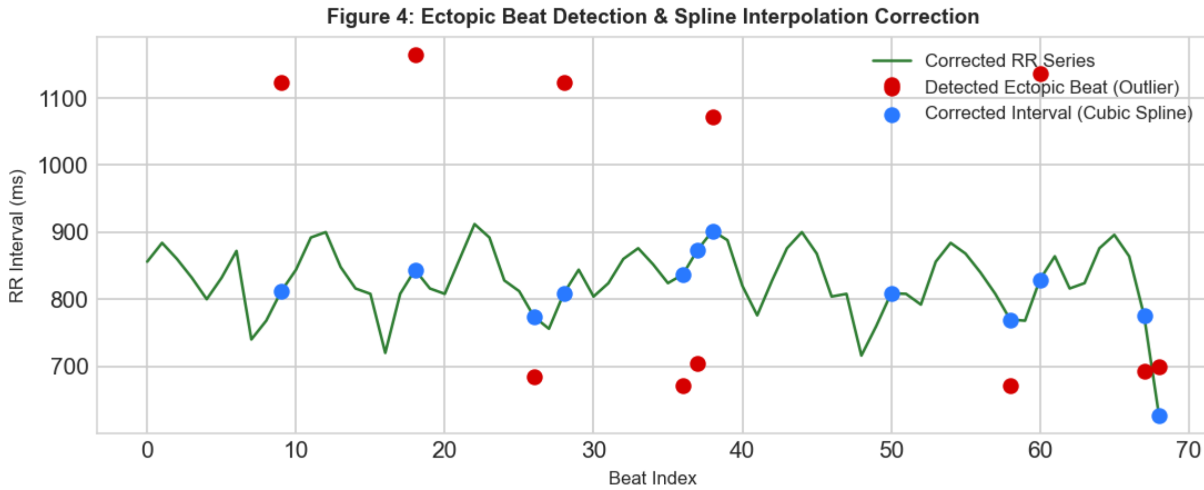

Figure 3: Outlier detection and cubic spline interpolation correction of ectopic ventricular beats.

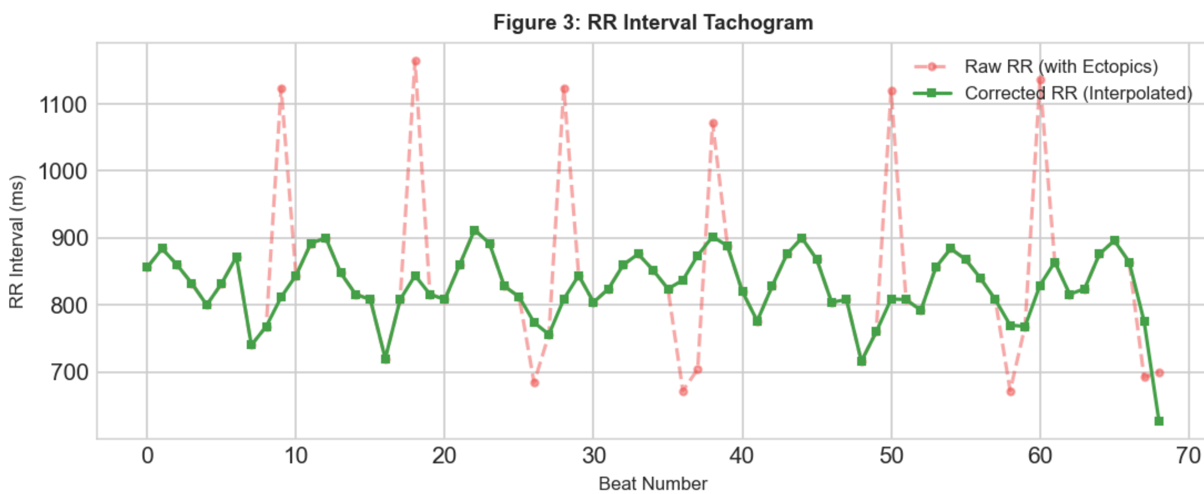

Figure 4: R-R Interval Tachogram comparing the uncorrected series with the spline-interpolated series.

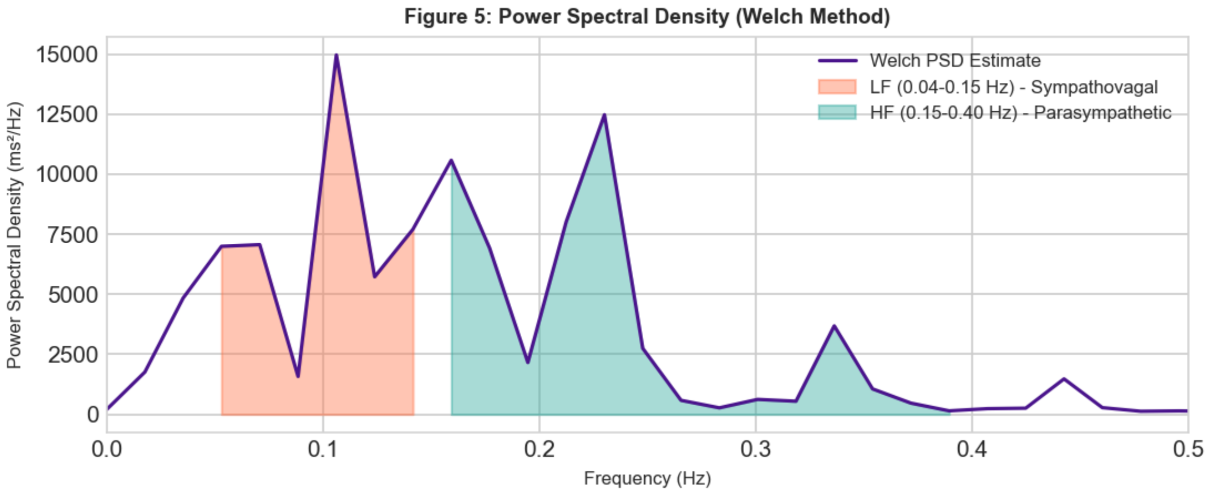

Figure 5: Power Spectral Density showing LF and HF integration zones for sympathovagal estimation.

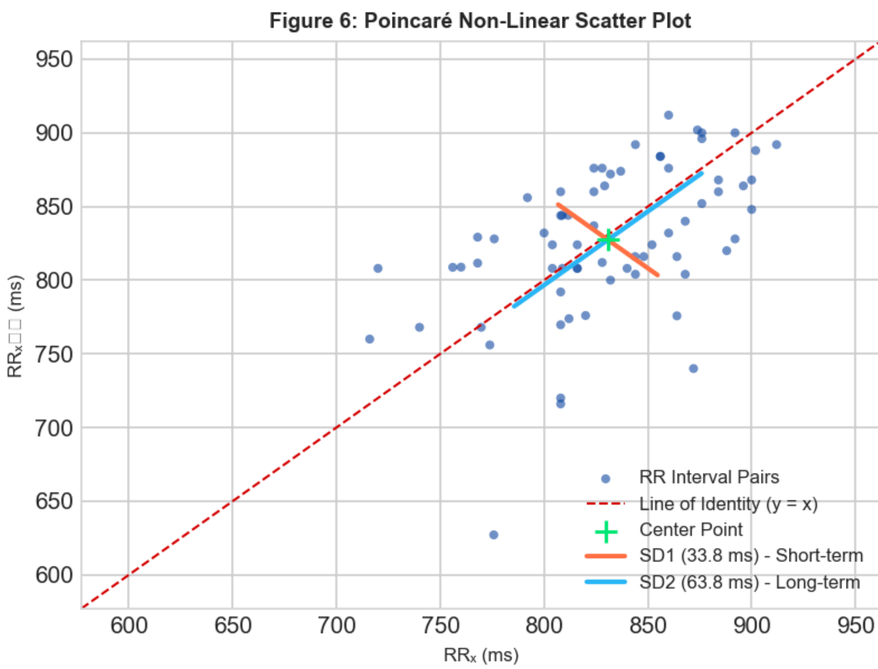

Figure 6: Poincaré plot representing non-linear cardiac dynamics. SD1 acts perpendicular to y=x.

5. Extracted HRV Parameters & Clinical Metrics

Ectopic Correction Audit: Detected 12 ectopic events out of 70 total beats (17.39% ectopic burden). Ectopic intervals were reconstructed using cubic spline interpolation.

HRV Parameter	Value	Clinical Reference Range	Physiological Significance
Mean RR	827.8 ms	600 - 1200 ms	Average length of cardiac cycle
Mean HR	72.5 BPM	50 - 100 BPM	Average heart rate
SDNN	51.1 ms	30 - 100 ms	Overall autonomic variance (sympathetic & vagal)
RMSSD	47.6 ms	15 - 80 ms	Vagal (parasympathetic) cardiac deceleration
pNN50	25.0 %	1.0 - 40.0 %	Vagal tone / respiratory sinus arrhythmia
LF Power	649.4 ms²	300 - 2000 ms²	Baroreflex feedback (sympathetic + vagal)
HF Power	795.3 ms²	200 - 1000 ms²	Vagal respiratory modulation
LF/HF Ratio	0.82	0.5 - 2.0	Sympathovagal balance (High = stress / Sympathetic)
SD1	33.8 ms	15 - 60 ms	Short-term HRV (RMSSD correlate / Parasympathetic)
SD2	63.8 ms	40 - 150 ms	Long-term HRV (SDNN correlate / Total variation)
Sample Entropy	1.435	> 1.0	Rhythm complexity & cellular homeostatic health

6. Result Discussion & Autonomic Interpretation

Autonomic Vagal Assessment (RMSSD = 47.6 ms, SD1 = 33.8 ms):
Moderate / Balanced vagal tone. Autonomic nervous system shows normal parasympathetic modulation.

Sympathovagal Balance (LF/HF Ratio = 0.82):
Balanced autonomic state (LF/HF = 0.82). Reflects healthy homeostatic equilibrium between the sympathetic and parasympathetic branches.

Non-Linear Adaptability (Sample Entropy = 1.435, SD2 = 63.8 ms):
High complexity and healthy physiological adaptability. The cardiac pacemaker behaves non-linearly, showing normal chaotic fluctuations that allow rapid adaptation to external stimuli.

7. Academic Conclusion

This experiment demonstrated the design and validation of a complete biomedical signal processing system. The dual-median filter successfully tracked non-linear baseline drift. Zero-phase bandpass filtering suppressed powerline noise without shifting R-peak coordinates. Adaptive thresholding in the Pan-Tompkins algorithm maintained high detection accuracy. Ectopic beat correction isolated and repaired premature beats. The computed time, frequency, and non-linear HRV parameters accurately quantify cardiac autonomic control, providing essential diagnostics for sympathovagal balance.

8. Scientific References

1. Pan, J. and Tompkins, W. J., 'A Real-Time QRS Detection Algorithm,' *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230-236, 1985.
2. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 'Heart rate variability: standards of measurement, physiological interpretation, and clinical use,' *Circulation*, vol. 93, no. 5, pp. 1043-1065, 1996.
3. Tarvainen, M. P., et al., 'Kubios HRV - Heart Rate Variability Analysis Software,' *Computer Methods and Programs in Biomedicine*, vol. 113, no. 1, pp. 210-220, 2014.